

Assignment 2 Report

Problem 2: Indexing Documents via Hadoop

Document Frequency Computation

Mapper

Each Wikipedia article is treated as one document. The mapper tokenizes the document, applies preprocessing (lowercasing, stopword filtering, stemming, etc.), and emits each distinct term once per document.

Output format:

TERM<TAB>DF

Reducer

The reducer sums counts to obtain the number of documents in which each term appears (DF).

Pseudocode (Mapper/Reducer)

```
Mapper(docId, docText):
    terms = empty set
    for token in tokenize(docText):
        t = preprocess(token)
        if t valid:
            terms.add(t)
    for t in terms:
        emit(t, 1)
```

```
Reducer(term, counts):
    DF = sum(counts)
    emit(term, DF)
```

TF-IDF Computation

Setup

The top 100 DF terms from Part (a) are loaded from `top100_df.tsv` via Hadoop distributed cache. Each term's TF is combined with its DF to compute:

$$\text{SCORE} = \text{TF} \times \log(10000/\text{DF} + 1)$$

Output format:

ID<TAB>TERM<TAB>SCORE

Mapper

For each document, the mapper builds a small in-memory map (stripe) of term frequencies (TF) for only those 100 terms, then emits the computed score.

Reducer

Passthrough reducer (emits the mapper output).

Pseudocode (Mapper/Reducer)

```
Mapper(docId, docText):
    tf = empty map
    for token in tokenize(docText):
        t = preprocess(token)
        if t in top100:
            tf[t] += 1
    for each t in tf:
        score = tf[t] * log(10000/DF[t] + 1)
        emit(docId + "\t" + t, score)

Reducer(key, values):
    emit(key, value)    // passthrough
```

Preprocessing Details

Applied in the mapper before emitting terms: - lowercase tokens and strip leading/trailing apostrophes
- drop tokens containing digits
- drop tokens shorter than 3 characters
- remove stopwords (checked before and after stemming)
- remove markup artifacts: apo, categori, extern, http, link, quot, refer

Runtime Analysis (Full Dataset)

Execution environment: WSL/Ubuntu, full dataset extracted into Linux-native storage.

Measured runtimes: - **Problem 2(a)**: real 4m19.412s - **Problem 2(b)**: real 4m6.313s

Screenshots

Here are the screenshots of successful full runs and runtimes:

Problem 2(a)

Problem 2(b)

Additional Notes

- The full dataset `Wikipedia-EN-20120601_ARTICLES.tar.gz` was used for final experiments.
- Sample outputs and full result files are included in `problem_2/results/`.

```

jack@Manzil-Laptop: /mnt/c/
FILE: Number of large read operations=0
FILE: Number of write operations=0
Map-Reduce Framework
  Map input records=10000
  Map output records=5805828
  Map output bytes=64068330
  Map output materialized bytes=75739989
  Input split bytes=1386997
  Combine input records=5805828
  Combine output records=5805828
  Reduce input groups=479009
  Reduce shuffle bytes=75739989
  Reduce input records=5805828
  Reduce output records=479009
  Spilled Records=11611656
  Shuffled Maps =10000
  Failed Shuffles=0
  Merged Map outputs=10000
  GC time elapsed (ms)=33163
  Total committed heap usage (bytes)=19903962677248
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=166222646
File Output Format Counters
  Bytes Written=5616472
real    4m19.412s
user    13m47.771s
sys     1m58.228s
jack@Manzil-Laptop: /mnt/c/Users/manzil/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2$ |

```

Figure 1: Problem 2(a) runtime screenshot

```

jack@Manzil-Laptop: /mnt/c/
FILE: Number of large read operations=0
FILE: Number of write operations=0
Map-Reduce Framework
  Map input records=10000
  Map output records=509190
  Map output bytes=10012479
  Map output materialized bytes=11490859
  Input split bytes=1386997
  Combine input records=0
  Combine output records=0
  Reduce input groups=509190
  Reduce shuffle bytes=11490859
  Reduce input records=509190
  Reduce output records=509190
  Spilled Records=1018380
  Shuffled Maps =10000
  Failed Shuffles=0
  Merged Map outputs=10000
  GC time elapsed (ms)=23221
  Total committed heap usage (bytes)=19903962677248
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=166222646
File Output Format Counters
  Bytes Written=15928164
real    4m6.313s
user    12m50.677s
sys     1m50.412s
jack@Manzil-Laptop: /mnt/c/Users/manzil/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2$ |

```

Figure 2: Problem 2(b) runtime screenshot

Verification (Optional)

An optional full-correctness verifier is available in `problem_2/src/VerifyTfIdf.java`. It recomputes every TF-IDF score from the dataset and compares it against `problem_2/results/tfidf_scores_full.tsv`. Compile/run commands are documented in `problem_2/problem_2.txt`.

```
jack@Manzil-Laptop: /mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2$ cd /mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2/problem_2
export JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64
javac -cp ".:/mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2/lib/opencv-tools-1.9.3.jar" \
src/VerifyTfIdf.java \
src/TextProcessingUtils.java
jack@Manzil-Laptop: /mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2/problem_2$ java -cp ".:/mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2/lib/opencv-tools-1.9.3.jar:src" \
VerifyTfIdf \
/home/jack/assignment2-data/Wikipedia-EN-20120601_ARTICLES \
/mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2/problem_2/results/top100_df.tsv \
/mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2/problem_2/results/tfidf_scores_full.tsv \
/mnt/c/Users/manzi/Downloads/NoSQL-main/NoSQL-main/NoSQL/Assignment2/data/stopwords.txt
Verification succeeded.
Docs processed: 10000
Matched: 509190
Mismatched: 0
Missing: 0
Extra output lines: 0
```

Figure 3: Problem 2(b) full verification screenshot

Verification Screenshot